Course Name: Computer Networks
Course Code: T7908
Faculty: Engineering
Course Credit: 3
Course Level: 4
Sub-Committee (Specialization): Computer Science
Learning Objectives:

The student will be able to:

Primary objective of this course is to get familiarly the basics of networking, Internet, Networking protocols and layered architecture of network

To explore and learn application layer protocols such as Domain Name Server, HTTP, FTP and SMTP

To explore transport layer protocols TCP and UDP. Further investigation of TCP flow control, congestion control and congestion avoidance

To explore and learn the basics of network layer, network layer protocol i.e. Internet Protocol (IP) and routing protocols and algorithms for wired and wireless network.

To learn error detection, error correction algorithms and algorithms related to reliable transmission.

To study various IEEE standards for wired networks and wireless networks.

Books Recommended:

Book	Author	Publisher
An Engineering Approach to Computer Networking	Keshav S.	Pearson Education, ISBN 981-235-986-9
Computer Networks, A 4th Edition	Tanenbaum	, PHI ISBP 81 - 203 - 2175 -8
Data Communications and Networking, 3rd edition	Fourauzan B.	Tata McGraw Hill Publications, 2004, ISBN 0 - 07 -058408 - 7.
Network Security Essentials,3rd edition	William Stallings	PEARSON

Course Outline:

Sr. No.	Topic	Actual Teaching Hours	Contact Hours Equivalence
1	**Computer Networks and the Internet:** Internet basics, protocol, network edge and network core, performance measures: delay, packet loss and throughput	**5**	**5**
2	**Application Layer:** Principles of network applications, HTTP, HTTP Message format, web caching, File Transfer Protocol, SMTP, DNS basics, Peer-to-Peer file distribution, Socket programming with TCP and UDP	**8**	**10**
3	**Transport Layer:** Overview of transport layer in the Internet, Transport and Network layer relationship, UDP, UDP checksum, concept of reliable data transfer, Go-Back-N and Selective-Repeat, TCP flow and congestion control	**12**	**20**
4	**The Network Layer:**	**15**	**20**

	Virtual circuit and datagram network, Internet protocol, IPV4 addressing, IPv6, ICMP, routing, routing algorithms: Link-State and Distance-Vector routing algorithm, Routing in the Internet, RIP, OSPF and BGP		
5	**The Link layer:** Link layer basics, error detection and correction, parity, checksum and CRC, random access control protocols, taking turns protocols, DOCSIS, Link virtualization, Multiprotocol Label Switching	**5**	**10**
	Total	**45**	**65**

Pre Requisites:

Knowledge of Operating system, probability and C programming is essential.

Evaluation:

Assignment

Class test

Seminar

Pedagogy:

Lectures

Online resources: NPTEL and Coursera

Research Paper Discussions